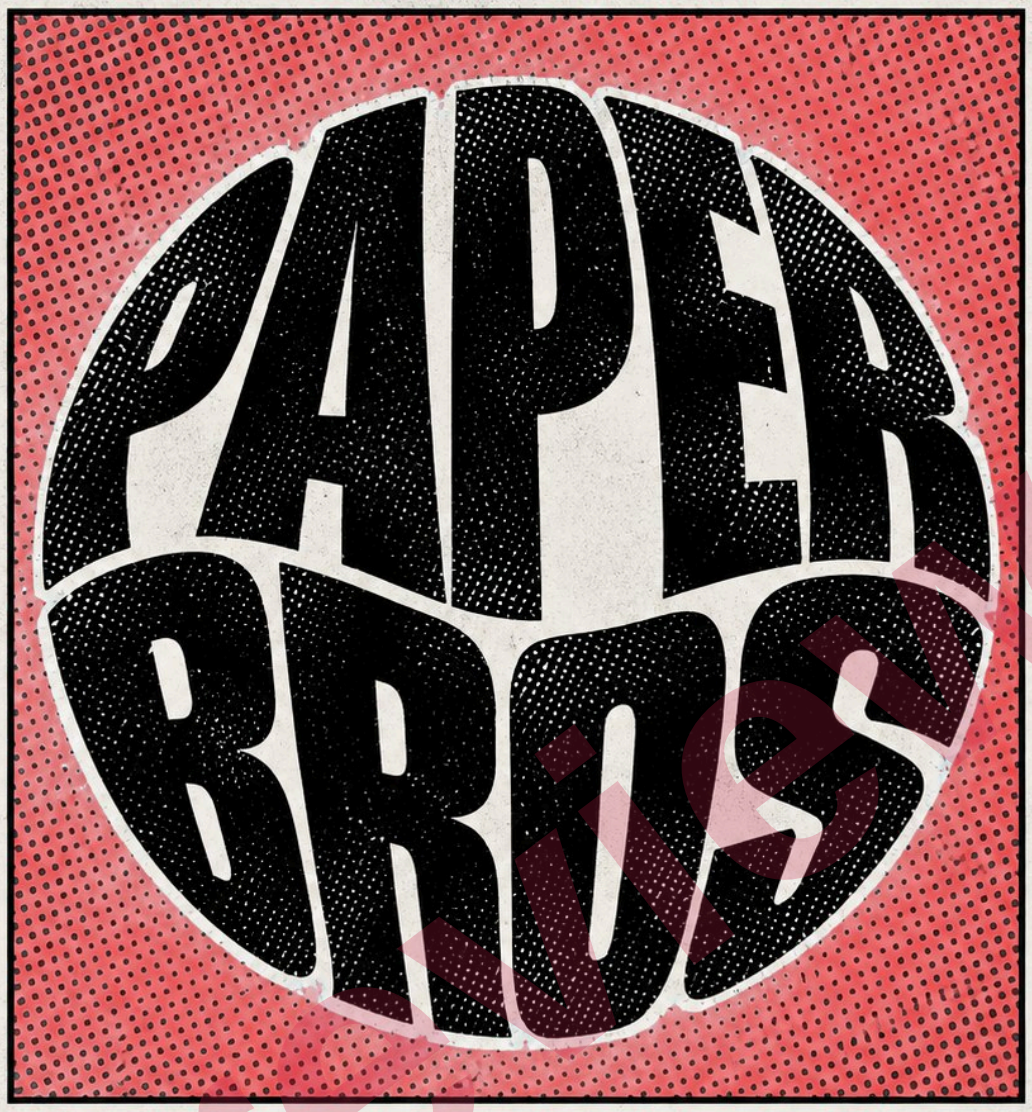
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A ONE STOP SOLUTION FOR YOUR BOARDS

DEVELOPED BY STUDENTS, FOR STUDENTS

O LEVEL CHEMISTRY 5070

PAPER 2 · THEORY

TOPICAL QUESTIONS + MARK SCHEME

2016 - 2026 · 388 QUESTIONS · 2026-2028 SYLLABUS



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CAMBRIDGE O LEVEL CHEMISTRY 5070

Paper 2 (Theory) · Topical questions and mark scheme, 2016 to May/June 2026

May/June and Oct/Nov, variants 21 and 22 · 42 papers · 388 questions in 1289 topic entries
Arranged by the 2026-2028 syllabus (5070) subtopics

CONTENTS

CODE	SUBTOPIC	QUESTIONS	QUESTION PAGE	MARK SCHEME PAGE
1	**STATES OF MATTER**	**37**	**4**	**605**
1.1	Solids, liquids and gases	27	4	605
1.2	Diffusion	10	16	607
2	**ATOMS, ELEMENTS AND COMPOUNDS**	**187**	**21**	**609**
2.1	Elements, compounds and mixtures	0	21	609
2.2	Atomic structure and the Periodic Table	36	21	609
2.3	Isotopes	13	37	611
2.4	Ions and ionic bonds	49	42	612
2.5	Simple molecules and covalent bonds	48	63	616
2.6	Giant covalent structures	19	85	620
2.7	Metallic bonding	22	95	621
3	**STOICHIOMETRY**	**182**	**103**	**624**
3.1	Formulae	70	103	624
3.2	Relative masses of atoms and molecules	23	122	628
3.3	The mole and the Avogadro constant	89	131	629
4	**ELECTROCHEMISTRY**	**53**	**175**	**638**
4.1	Electrolysis	44	175	638
4.2	Hydrogen-oxygen fuel cells	9	194	642
5	**CHEMICAL ENERGETICS**	**38**	**198**	**643**
5.1	Exothermic and endothermic reactions	38	198	643
6	**CHEMICAL REACTIONS**	**141**	**225**	**648**
6.1	Physical and chemical changes	7	225	648
6.2	Rate of reaction	34	227	648
6.3	Reversible reactions and equilibrium	54	250	653
6.4	Redox	46	280	658
7	**ACIDS, BASES AND SALTS**	**94**	**299**	**662**
7.1	The characteristic properties of acids and bases	48	299	662
7.2	Oxides	12	316	665
7.3	Preparation of salts	34	321	665
8	**THE PERIODIC TABLE**	**66**	**338**	**669**
8.1	Arrangement of elements	15	338	669
8.2	Group I properties	6	345	670
8.3	Group VII properties	17	347	670
8.4	Transition elements	24	354	671
8.5	Noble gases	4	361	673
9	**METALS**	**79**	**364**	**675**
9.1	Properties of metals	8	364	675
9.2	Uses of metals	7	366	675
9.3	Alloys and their properties	11	368	676
9.4	Reactivity series	22	372	677
9.5	Corrosion of metals	11	382	678
9.6	Extraction of metals	20	385	679
10	**CHEMISTRY OF THE ENVIRONMENT**	**78**	**396**	**682**
10.1	Water	12	396	682
10.2	Fertilisers	13	402	682
10.3	Air quality and climate	53	407	683

Preview

PREVIEW

CODE	SUBTOPIC	QUESTIONS	QUESTION PAGE	MARK SCHEME PAGE
11	**ORGANIC CHEMISTRY**	**245**	**431**	**688**
11.1	Formulae, functional groups and terminology	61	431	688
11.2	Naming organic compounds	6	463	693
11.3	Fuels	22	465	693
11.4	Alkanes	19	474	695
11.5	Alkenes	34	483	697
11.6	Alcohols	24	499	699
11.7	Carboxylic acids	39	510	701
11.8	Polymers	40	534	705
12	**EXPERIMENTAL TECHNIQUES AND CHEMICAL ANALYSIS**	**89**	**569**	**713**
12.1	Experimental design	2	569	713
12.2	Acid-base titrations	13	569	713
12.3	Chromatography	2	577	714
12.4	Separation and purification	5	579	714
12.5	Identification of ions and gases	67	580	714

Preview

1 STATES OF MATTER

UNIT 1

1.1 Solids, liquids and gases

27 QUESTIONS

Q1 **2016/May-June/22/QA3 (b)**

A3 Esters are used as food flavourings and solvents.

(b) Ethyl ethanoate evaporates at room temperature.

(i) What is meant by the term *evaporation*?

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..... [1]

(ii) A sample of ethyl ethanoate in a beaker is moved into a colder room.

Explain, in terms of the kinetic particle theory, why this results in a decrease in the rate of evaporation.

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..... [2]

(iii) The table shows some information about different esters.

name	structure	relative molecular mass (M_r)
methyl ethanoate	$CH_3CO_2CH_3$	74
ethyl ethanoate	$CH_3CO_2C_2H_5$	88
propyl ethanoate	$CH_3CO_2C_3H_7$	102
butyl ethanoate	$CH_3CO_2C_4H_9$	116
pentyl ethanoate	$CH_3CO_2C_5H_{11}$	130

Which ester has the **lowest** rate of evaporation at room temperature and pressure?

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Explain your answer.

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[2]

Preview

Q2 **2017/May-June/21/QB10 (d)**

B10 The table shows some information about the homologous series of unbranched carboxylic acids.

name	structure	boiling point/°C
methanoic acid	HCO_2H	101
ethanoic acid	CH_3CO_2H	118
propanoic acid	$CH_3CH_2CO_2H$	141
butanoic acid	$CH_3CH_2CH_2CO_2H$	164
pentanoic acid	$CH_3CH_2CH_2CH_2CO_2H$	186

(d) Ethanoic acid is a liquid at room temperature and has a boiling point of 118 °C.

Describe the changes in both the arrangement and movement of the molecules when ethanoic acid is heated from room temperature to 120 °C.

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.....[3]

Q3 **2017/May-June/22/QB10 (d)**

B10 The table shows some information about the homologous series of unbranched alcohols.

name	structure	boiling point / °C
methanol	CH_3OH	65
ethanol	CH_3CH_2OH	79
propanol	$CH_3CH_2CH_2OH$	97
butanol	$CH_3CH_2CH_2CH_2OH$	117
pentanol	$CH_3CH_2CH_2CH_2CH_2OH$	138

(d) Ethanol is a gas at 100 °C.

Describe the changes in the arrangement and movement of the molecules when ethanol is cooled from 100 °C to 25 °C.

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Preview